
Where Does Platonic Convergence Live? A Linear Readout Gains Where the Galaxy Embedding Bends Toward the Label and Loses Where It Bends Away

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by
2 measuring the convergence of foundation-model representations of galaxy images
3 with mutual k -nearest-neighbour agreement (MKNN). We ask whether that conver-
4 gence is concentrated where the embedding manifold is flat. A pointwise curvature
5 field from a trained decoder, validated against analytic truth, finds mean-curvature
6 magnitude correlating negatively with cross-survey neighbour agreement at every
7 latent dimension, and local density removes most, half or little of it ($d = 20, 25,$
8 32). A second-order expansion of a linear map restricted to the manifold explains
9 why: a readout sees the second fundamental form contracted with its normal com-
10 ponent, $\langle w_N, \Pi \rangle$, an alignment’s second-order error is a connection mismatch plus
11 the relative second fundamental form, and mean curvature retains only the metric
12 trace of Π and determines neither. On 86,471 labelled galaxies a linear readout
13 gains accuracy where the manifold bends toward the label’s own curvature and
14 loses it where the manifold bends away. Intervening on the readout at a fixed man-
15 ifold, the decoder’s second-order term raises local R^2 at 79–99% of anchors, its
16 mirror image lowers it at 97–100%, and a random normal direction of the same
17 norm does neither, on five encoders; across anchors, the mismatch between the
18 label Hessian and the probe-facing Hessian is negative wherever the label has lo-
19 cal structure and survives cross-fitting and a change of encoder. Mean curvature
20 of a single representation is therefore not a viable explanation for weak Platonic
21 convergence; the obstruction is relative and, for the fitted global linear alignment,
22 already present at first order.

23 1 Introduction

24 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data
25 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on the sky:
26 galaxies imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC) share only 0.3 to
27 2.3 percent of nearest neighbours across surveys under a family of vision models. We ask where on
28 the manifold that agreement lives: if two models disagree most where the embedding surface bends
29 sharply, local mean curvature should associate negatively with local cross-survey agreement.

30 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
31 trained decoder, validated against analytic truth in the production regime; (ii) a density-controlled
32 test of the curvature–alignment prediction, which finds no effect free of density; (iii) the residual
33 expansion of a linear map on a submanifold, which names the curvature such a map responds to
34 and shows mean-curvature magnitude is blind to it; and (iv) known-surface, cross-encoder and inter-
35 ventional tests on the readout side: a linear readout gains local accuracy where the manifold bends

36 toward the label’s curvature and loses it where the manifold bends away, at nearly every anchor on
 37 five encoders.

38 2 Pointwise curvature from a decoder

39 **Data.** DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both
 40 HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019], (Platonic Universe release [Du-
 41 raphe et al., 2025]); we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-
 42 survey ViT-B embeddings with catalogue labels (r -band magnitude, photometric redshift, smooth
 43 fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the readout experiment.
 44 Embeddings are unit-normalized (data on S^{767}). Curvature is evaluated only where J has full rank.

45 **Instrument.** Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden lay-
 46 ers of width 250, SiLU), fit on 80% of rows; curvature is evaluated on held-out rows. With
 47 $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental
 48 form $\Pi = P_N D^2 F(z)$, the mean-curvature vector of the raw decoder surface is $H_{\text{raw}} = \text{tr}_g \Pi$, by
 49 automatic differentiation. On the unit sphere every d -submanifold has radial mean curvature $-d$,
 50 which says nothing about shape; differentiating $F/\|F\|$ instead splits H exactly as $-d\hat{x} + H^S$
 51 ($H_{\text{rad}} = -d$ to 10^{-14}), and H^S carries the shape. Section 3 uses the pre-registered $\|H_{\text{raw}}\|$; Sec-
 52 tions 4–5 use the projected decoder. Decoder curvature has been a regularizer [Lee and Park, 2023]
 53 and an estimator [Acosta et al., 2023].

54 **Validation.** Against closed-form curvature, at $D = 768$ an in-sphere surface $G(z) =$
 55 $Q \text{ normalize}([\phi(z); 0.8h(z); 0 \dots])$ (ϕ inverse stereographic $\mathbb{R}^{16} \rightarrow S^{16}$, h a scalar sum of four
 56 Gaussian bumps, Q a rotation; 18-dimensional span) at $n = 86,471$, $k = 2,048$ gives rank 0.94
 57 against the true $\|H^S\|$, direction cosine 0.999 and magnitude ratio 1.00 for the full vector.

58 3 Curvature and cross-survey convergence

59 **Probe.** Per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k = 20$
 60 between the HSC and Legacy Survey embeddings, correlated with $\|H_{\text{raw}}\|$ over all 10,000 points at
 61 $d \in \{20, 25, 32\}$.

62 **Result.** The raw association $\rho(\|H_{\text{raw}}\|, \text{MKNN})$ is -0.112 , -0.128 and -0.024 at $d = 20, 25, 32$
 63 ($p < 10^{-5}$, $< 10^{-5}$, 0.009). But k -NN statistics are density-sensitive [Radovanović et al., 2010]:
 64 local density ($k = 30$ estimate on the Legacy embedding) correlates with agreement at -0.21 at
 65 every d and with $\|H_{\text{raw}}\|$ at $+0.43$, $+0.32$ and $+0.01$, so residualizing on density leaves -0.024 ,
 66 -0.066 and -0.022 (78%, 49% and 8% removed). Under a permutation null within density strata
 67 only $d = 25$ survives ($p < 5 \times 10^{-5}$; three decoder seeds; CKA [Kornblith et al., 2019] agrees), and
 68 removing the radial component of H_{raw} collapses it to -0.023 .

69 4 What a linear map sees on a curved manifold

70 Let $M \subset \mathbb{R}^D$ be a d -dimensional submanifold, $d < D$; at x_0 take an orthonormal tangent frame J ,
 71 the second fundamental form Π (normal-valued, symmetric $d \times d$), $H = \text{tr} \Pi$, and for a vector w its
 72 tangent and normal parts $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) +$
 73 $O(|u|^3)$ [Lee, 2018], so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

74 For $M \subset S^{D-1}$ the normal bundle splits into the sphere’s radial direction and the rest, $\Pi = \Pi^S - g\hat{x}$,
 75 so $\text{Hess}_M(w \cdot x) = \langle w_N, \Pi^S \rangle - (w \cdot \hat{x})g$ (2): a linear readout of a unit-normalized embedding is bent
 76 by the manifold’s shape and, whatever the manifold, by the sphere in proportion to the readout’s own
 77 value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and a patch $u \sim N(0, s^2 I_d)$
 78 in normal coordinates, embedding and label locally C^4 (the $\frac{1}{2}$ is the Gaussian constant; real k -NN
 79 patches share only the tensor structure), with $\bar{c}$ the patch-mean residual, $\delta = [\nabla_M y]_J - J^\top w$, $\Delta =$

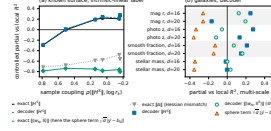


Figure 1: (a) Known surface, intrinsically linear label, exact geometry: three-control partial against local R^2 as the density-curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$, multi-scale control: the decoder's $\|H^S\|$ (filled squares), shape term (open circles), sphere term (open triangles).

label	$\ H^S\ $		shape $\ \langle w_N, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ $		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control; alignment is $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$. * not significant at 0.05; the three-control partials agree in sign in every significant cell.

80 $\text{Hess}_M y - \langle w_N, \Pi \rangle,$

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2\|\delta\|^2 + \frac{1}{2}s^4\|\Delta\|_F^2 + O(s^4\|\delta\|) + O(s^6). \quad (3)$$

81 For two embeddings F, G of the same objects in a common chart u and a linear map A with
82 $AJ_F = J_G$, the same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2}[J_G(\Gamma^G - \Gamma^F) + \Pi_G -$
83 $A\Pi_F](u, u) + O(|u|^3)$, with Γ the induced connections: the second-order obstruction to linear
84 alignment is a connection mismatch plus the *relative* second fundamental form; the extrinsic term's
85 normal part is $\Pi_G - P_N^G A\Pi_F$.

86 So $\|H\|$, the norm of the metric trace of Π , need not correlate with either $\langle w_N, \Pi \rangle$ or the relative term;
87 the second-order term scales as s^4 and, with fixed- k neighbourhoods, density sets s , so curvature
88 effects scored on patches are density-sensitive.

89 5 Probe-facing curvature: bending toward the label helps, bending away 90 hurts

91 Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at the pre-registered $\alpha = 100$
92 from embedding to label; local R^2 inside each of 512 anchors' 2,048-neighbourhoods; rank-partial
93 Spearman against local R^2 with Freedman-Lane p -values [Freedman and Lane, 1983], control-
94 ling for log patch radius, local label variance and count and, stronger, log radius at five scales
95 $k \in \{16, \dots, 2048\}$. Per anchor we compute $\|H^S\|$ and $\|\langle w_N, \Pi \rangle\|_g$, split by Eq. (2) into shape
96 and sphere terms.

97 **Known surface.** The validation surface resampled with weight $\|H^S\|^\gamma$, so only where the points
98 sit changes (Figure 1a): the $\|H^S\|$ partial follows the sampling from -0.29 to $+0.22$, reproduced by
99 the decoder, while the probe-facing Hessian $\|\langle w_N, \Pi \rangle\|$ holds at -0.74 to -0.80 at every sampling,
100 there entirely the sphere term of Eq. (2).

101 **Galaxies.** Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing
102 Hessian (rank 0.96–0.99 with the full norm). We estimate $\text{Hess}_M y$ per anchor by a least-squares
103 quadratic of the label on the tangent-projected coordinates $u = g^{-1}J^\Gamma(x - x_0)$ of its 2,048 neigh-
104 bours (a Monge chart: the quadratic coefficient is the covariant Hessian). Three regularities (Table 1;
105 Appendix A for other seeds and widths). The sphere term is negative on every label and both d
106 (-0.13 to -0.36) under the sealed probe; with a weak-ridge probe ($\alpha = 1$, Appendix B) it stays neg-
107 ative for magnitude and smooth fraction (-0.21 , -0.12) and vanishes for redshift and stellar mass,
108 where the sealed-probe sign came with ridge shrinkage of extreme predictions. Wherever the label

has local structure (magnitude, redshift, smooth fraction; stellar mass is indistinguishable from zero throughout) the mismatch is negative (-0.09 to -0.45 ; unchanged under cross-fitting, Appendix B) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive ($+0.07$ to $+0.43$). The shape term alone is label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift): with $K = \langle w_N, \Pi \rangle$ it is one of three terms in $\|\Delta\|^2 = \|\text{Hess}_M y\|^2 - 2\langle \text{Hess}_M y, K \rangle + \|K\|^2$, so raising it shrinks or enlarges the mismatch with the sign of the cross term. The same regression on the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$ (median cosine with the decoder’s 0.55 – 0.65).

Bending toward helps, bending away hurts. By Eq. (3) curvature lowers the second-order residual relative to a flat patch iff $2\langle \text{Hess}_M y, K \rangle > \|K\|^2$: the manifold must bend toward the label’s own curvature, and not too far. (The optimal normal weight of least squares on a hypersurface, label second derivatives contracted with principal curvatures over their squares, was derived as a uniqueness and stability condition [Liu et al., 2023]; here it is measured per anchor on learned embeddings. Manifold capacity [Chung et al., 2016, Slatton et al., 2026] uses dimension and radius, flattening [Psenka et al., 2024] treats curvature as an obstacle, latent curvature tracks generalization [Kaufman and Azencot, 2023] and biases local linear regression [Cheng and Wu, 2013]; none conditions on the sign of $\langle \text{Hess}_M y, K \rangle$.) The partials compare anchors, and no anchor is flat, so we test the condition by intervening on the readout at a fixed manifold. At each anchor the fitted weight’s in-sphere normal part is replaced by t times the decoder’s second-order term $\frac{1}{2}\langle w_S, \Pi^S \rangle(u, u)$: $t = 1$ is the manifold’s own bending, $t = -1$ its mirror image, $t = 0$ flat. On five encoders and all four labels, $t = 1$ beats flat at 79–99% of anchors (median local R^2 gain $+0.013$ to $+0.031$), $t = -1$ is worse than flat at 97–100% (-0.025 to -0.055), and a random normal direction of the same norm does neither (Appendix E). t^* sits at 1.5 to 3.6: the ridge-shrunk readout under-uses the bending, as the weak-ridge gain above suggests.

Across encoders. The quadratic-chart estimator’s mean-bending trace at the same anchors of five encoders (ViT-B, DINOv3, CLIP-B, ConvNeXt-B, ViT-L; $d = 16$) has density-controlled associations with the magnitude probe’s local R^2 of both signs (-0.24 to $+0.27$), as Eq. (1) allows. The decoder test repeated per encoder (Appendix D) is encoder-stable where the expansion says it should be: the mismatch partial is negative for magnitude and redshift on all five encoders (-0.32 to -0.61 ; cross-fitted -0.28 to -0.62) and the alignment positive for magnitude on all five ($+0.13$ to $+0.53$) and for redshift on four; the shape term’s sign varies with encoder as well as label; stellar mass stays null but for one mismatch cell (-0.12).

6 Discussion

A linear readout of a galaxy embedding gains accuracy where the manifold bends toward the label’s curvature and loses it where the manifold bends away, at nearly every anchor and on five encoders. The residual expansion names the quantity, $\langle w_N, \Pi \rangle$ against $\text{Hess}_M y$, and a decoder’s pointwise second fundamental form measures it at production scale, agreeing with analytic truth and with the data. For the linear alignment map that recent accounts rest alignment on [Bangachev et al., 2026], the relevant second-order quantity is not one embedding’s mean curvature but $\Pi_G - P_N^G A \Pi_F$, which needs both second fundamental forms and a map between tangent frames. Our pilot (Appendix C) finds a 44% first-order tangent-frame residual under the fitted global ridge map and no estimator-stable second-order association. On this survey pair, single-model mean curvature has no robust density-controlled association with cross-survey convergence.

Limitations. One survey pair. The label’s Hessian is a local estimate (R^2 gain over linear 0.10 – 0.17); its norm is reliably ranked (split-half ρ 0.92) but its direction only moderately (tensor cosine 0.19 – 0.35). On a synthetic surface with known curvature the mismatch partial reproduces (-0.14 to -0.88) but the alignment partial has no stable sign; Appendix E, not the fixture, ties alignment to local R^2 . Stars are per cell; mismatch and alignment are the primary hypotheses, the rest exploratory. Permutation p -values assume anchor exchangeability under overlapping patches.

References

- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL <https://arxiv.org/abs/2405.07987>.

- 161 Kshitij Duraphe, Michael J. Smith, Shashwat Sourav, and John F. Wu. The platonic universe: Do foundation
162 models see the same sky? In *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences*, 2025.
163 URL <https://arxiv.org/abs/2509.19453>.
- 164 Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, et al. DINOv3. *arXiv*
165 *preprint*, 2025. URL <https://arxiv.org/abs/2508.10104>.
- 166 Hiroaki Aihara, Nobuo Arimoto, Robert Armstrong, et al. The Hyper Suprime-Cam SSP Survey: Overview
167 and survey design. *Publications of the Astronomical Society of Japan*, 70(SP1):S4, 2018. doi: 10.1093/pasj/
168 psx066.
- 169 Arjun Dey, David J. Schlegel, Dustin Lang, et al. Overview of the DESI Legacy Imaging Surveys. *The Astro-*
170 *nomical Journal*, 157(5):168, 2019. doi: 10.3847/1538-3881/ab089d.
- 171 Michael J. Smith, Ryan J. Roberts, Eirini Angeloudi, and Marc Huertas-Company. AstroPT: Scaling large obser-
172 vation models for astronomy. *arXiv preprint*, 2024. URL <https://arxiv.org/abs/2405.14930>.
- 173 Yonghyeon Lee and Frank Chongwoo Park. On explicit curvature regularization in deep generative models. In
174 *2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG-ML)*, ICML, 2023.
175 URL <https://arxiv.org/abs/2309.10237>.
- 176 Francisco Acosta, Sophia Sanborn, Khanh Dao Duc, Manu Madhav, and Nina Miolane. Quantifying extrinsic
177 curvature in neural manifolds. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition*
178 *Workshops (CVPRW)*, pages 610–619, 2023. doi: 10.1109/CVPRW59228.2023.00068.
- 179 Miloš Radovanović, Alexandros Nanopoulos, and Mirjana Ivanović. Hubs in space: Popular nearest neighbors
180 in high-dimensional data. *Journal of Machine Learning Research*, 11:2487–2531, 2010.
- 181 Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network rep-
182 resentations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*,
183 2019.
- 184 John M. Lee. *Introduction to Riemannian Manifolds*, volume 176 of *Graduate Texts in Mathematics*. Springer,
185 2nd edition, 2018. doi: 10.1007/978-3-319-91755-9.
- 186 Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016.
187 arXiv:1610.01644.
- 188 David Freedman and David Lane. A nonstochastic interpretation of reported significance levels. *Journal of*
189 *Business & Economic Statistics*, 1(4):292–298, 1983. doi: 10.1080/07350015.1983.10509354.
- 190 Liangchen Liu, Juncai He, and Yen-Hsi Tsai. Linear regression on manifold structured data: the impact of
191 extrinsic geometry on solutions. In *Proceedings of the 2nd Annual Workshop on Topology, Algebra, and*
192 *Geometry in Machine Learning (TAG-ML)*, volume 221 of *Proceedings of Machine Learning Research*, pages
193 557–576. PMLR, 2023.
- 194 SueYeon Chung, Daniel D. Lee, and Haim Sompolskiy. Linear readout of object manifolds. *Physical Review*
195 *E*, 93(6):060301, 2016. doi: 10.1103/PhysRevE.93.060301.
- 196 Will Slatton, Chi-Ning Chou, and SueYeon Chung. Linear readout of neural manifolds with continuous vari-
197 ables, 2026. URL <https://arxiv.org/abs/2603.10956>.
- 198 Michael Psenka, Druv Pai, Vishal Raman, Shankar Sastry, and Yi Ma. Representation learning via manifold
199 flattening and reconstruction. *Journal of Machine Learning Research*, 25(132):1–47, 2024. URL <https://www.jmlr.org/papers/v25/23-0615.html>.
- 201 Ilya Kaufman and Omri Azencot. Data representations’ study of latent image manifolds. In *Proceedings of*
202 *the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning*
203 *Research*, pages 15928–15945. PMLR, 2023.
- 204 Ming-Yen Cheng and Hau-Tieng Wu. Local linear regression on manifolds and its geometric interpretation.
205 *Journal of the American Statistical Association*, 108(504):1421–1434, 2013. doi: 10.1080/01621459.2013.
206 827984.
- 207 Kiril Bangachev, Guy Bresler, and Yury Polyanskiy. Representation alignment rests on linear structure, 2026.
208 URL <https://arxiv.org/abs/2605.28870>.

Appendix A: decoder ablations

Table 2 repeats the four quantities of Table 1 at $d = 16$ under the multi-scale density control for two further decoder initialisation seeds and for a decoder of width 400^3 instead of 250^3 (all fits reach variance explained 0.952). Every significant sign of the main table is retained.

label	quantity	seed 0	seed 1	seed 2	400^3
mag_r	shape	+0.12	+0.05*	+0.12	+0.05*
	sphere	-0.26	-0.26	-0.27	-0.26
	mismatch	-0.39	-0.35	-0.34	-0.34
	alignment	+0.35	+0.31	+0.37	+0.41
photo_z	shape	-0.06*	-0.09*	-0.09	-0.06*
	sphere	-0.33	-0.35	-0.34	-0.35
	mismatch	-0.45	-0.43	-0.42	-0.41
	alignment	+0.26	+0.24	+0.25	+0.25
smooth_fraction	shape	+0.07*	+0.20	+0.15	+0.10
	sphere	-0.13	-0.15	-0.14	-0.11
	mismatch	-0.09*	-0.05*	-0.08*	-0.07*
	alignment	+0.19	+0.14	+0.09	+0.10
stellar_mass	shape	+0.00*	-0.06*	+0.05*	+0.09
	sphere	-0.35	-0.35	-0.35	-0.35
	mismatch	-0.04*	+0.03*	+0.02*	+0.03*
	alignment	+0.01*	-0.02*	+0.03*	-0.02*

Table 2: Decoder ablations, $d = 16$, 512 anchors, multi-scale density control. * not significant at 0.05.

Appendix B: sensitivity of the mismatch and sphere columns

Cross-fitting: each 2,048-patch is split at random into halves; $\text{Hess}_M y$ is fitted on one half and local R^2 scored on the other (both directions shown), so the Hessian estimate and the outcome share no rows. The split-half cosine between the two Hessian estimates is the reliability of the estimate. Weak ridge: the probe refit at $\alpha = 1$ instead of 100 (global out-of-sample R^2 rises by 0.10 to 0.15), which substantially reduces the shrinkage of extreme predictions; the sphere term is recomputed from that probe.

label	d	mismatch, cross-fit		alignment, cross-fit		Hess. split cos	sphere, $\alpha = 1$
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	p50	(main: $\alpha = 100$)
mag_r	16	-0.38	-0.34	+0.33	+0.31	+0.35	-0.21 (-0.26)
mag_r	20	-0.43	-0.40	+0.39	+0.37	+0.24	-0.22 (-0.26)
photo_z	16	-0.42	-0.36	+0.23	+0.20	+0.32	+0.05* (-0.33)
photo_z	20	-0.42	-0.42	+0.22	+0.22	+0.20	+0.03* (-0.36)
smooth_fraction	16	-0.10	-0.17	+0.17	+0.19	+0.33	-0.12 (-0.13)
smooth_fraction	20	-0.15	-0.21	+0.05*	+0.03*	+0.23	-0.13 (-0.15)
stellar_mass	16	+0.01*	-0.04*	-0.03*	+0.06*	+0.31	-0.02* (-0.35)
stellar_mass	20	-0.06*	-0.07*	+0.01*	-0.00*	+0.19	-0.02* (-0.35)

Table 3: Cross-fitted mismatch and alignment partials (multi-scale control), split-half reliability of the label Hessian, and the sphere-term partial under a weak-ridge probe.

Appendix C: relative second fundamental form, pilot

Two sphere-projected decoders (HSC = F , Legacy = G ; Phase 7 protocol), a global ridge map A from x_F to x_G (fit on the 8,000 training rows), 2,048 seeded anchors. Columns: holdout R^2 of A ; median first-order obstruction $\|AJ_F - J_GL\|/\|AJ_F\|$ with $L = J_G^+ AJ_F$; then multi-scale density-controlled partials against MKNN ($k = 20$; log radius at $k \in \{10, 30, 100, 300\}$ in both spaces) of each decoder's $\|H^S\|$, the first-order obstruction, $\|\Pi_G(L, L) - P_N^G A \Pi_F\|$, the same with L fitted

on 256 neighbours' latent codes, and a decoder-free estimate from the quadratic coefficient of the alignment residual on F 's tangent coordinates.

run	R_A^2	tan. resid	$\ H^S\ $ HSC	$\ H^S\ $ Legacy	tan. resid	Π_{rel}	Π_{rel} (local L)	Π_{rel} (data)
$d = 20$, seed 0	0.64	0.44	-0.15	+0.04*	-0.18	+0.04*	-0.18	+0.13
$d = 25$, seed 0	0.64	0.43	-0.10	+0.06	-0.16	+0.05	-0.23	+0.13
$d = 20$, seed 1	0.64	0.44	-0.11	+0.05	-0.19	+0.04	-0.20	+0.14

Table 4: Relative-II pilot. * not significant at 0.05.

Appendix D: the same test on four further encoders

Table 5 repeats Table 1 at $d = 16$ on the same 86,471 galaxies embedded by four further encoders from the same release (DINOv3 ViT-B/16, CLIP ViT-B, ConvNeXt-B, ViT-L; widths 768, 512, 1,024, 1,024; each unit-normalized), with a decoder of the same width and protocol fitted per encoder (seed 0; variance explained 0.966, 0.985, 0.965, 0.968 for the four new fits), 512 anchors, $k = 2,048$, multi-scale density control. The first row per label is the probe's global out-of-sample R^2 ; Table 6 gives the cross-fitted mismatch and alignment and the split-half reliability of the label Hessian per encoder.

label	quantity	ViT-B	DINOv3	CLIP-B	ConvNeXt-B	ViT-L
mag_r	global R^2	0.52	0.57	0.50	0.52	0.52
	shape	+0.12	-0.11	+0.16	+0.16	-0.21
	sphere	-0.26	-0.23	-0.40	-0.24	-0.08*
	mismatch	-0.39	-0.57	-0.34	-0.32	-0.44
	alignment	+0.35	+0.53	+0.32	+0.21	+0.13
photo_z	global R^2	0.51	0.56	0.49	0.49	0.51
	shape	-0.06*	-0.05*	-0.17	+0.29	-0.16
	sphere	-0.33	-0.21	-0.29	-0.46	-0.44
	mismatch	-0.45	-0.61	-0.45	-0.48	-0.46
	alignment	+0.26	+0.33	+0.06*	+0.22	+0.14
smooth_fraction	global R^2	0.53	0.59	0.49	0.56	0.55
	shape	+0.07*	-0.08*	+0.20	+0.24	+0.06*
	sphere	-0.13	-0.06*	+0.05*	-0.15	-0.11
	mismatch	-0.09*	-0.33	-0.25	-0.20	-0.01*
	alignment	+0.19	-0.02*	+0.24	+0.04*	+0.14
stellar_mass	global R^2	0.48	0.54	0.50	0.49	0.49
	shape	+0.00*	-0.15	-0.10	-0.11	-0.13
	sphere	-0.35	-0.26	-0.38	-0.40	-0.38
	mismatch	-0.04*	+0.03*	-0.12	-0.06*	-0.02*
	alignment	+0.01*	-0.05*	-0.01*	+0.07*	-0.03*

Table 5: Cross-encoder replication, $d = 16$, multi-scale density control. * not significant at 0.05.

Appendix E: intervening on the readout's normal component

The partials of Table 1 compare anchors with one another and cannot say whether bending away from the label's curvature hurts relative to no bending, since no anchor is flat. Here the readout is changed at a fixed manifold. At each anchor the fitted ridge weight is split into tangent, radial and in-sphere normal parts $w = w_T + w_{\text{rad}} + w_S$, and the readout $w_T + w_{\text{rad}} + t q$ is scored on the anchor's 2,048 neighbours with the local intercept refit, where q is the decoder's second-order term $\frac{1}{2}\langle w_S, \Pi^S \rangle(u, u)$ in the anchor's chart coordinates: $t = 1$ is the manifold bending as it does in the probe's normal direction, $t = -1$ the mirror-image bending, $t = 0$ flat. The local sum of squares is an exact quadratic in t with minimum t^* . Columns: fraction of anchors where $t = 1$ beats flat (help) and where $t = -1$ is worse than flat (hurt), the median change in local R^2 at $t = \pm 1$, the median t^* ; then the same for a random in-sphere normal direction of the same norm as w_S .

encoder	label	mismatch, cross-fit		alignment, cross-fit		Hess. split cos p50
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	
DINOv3	mag_r	−0.60	−0.62	+0.50	+0.50	+0.35
	photo_z	−0.56	−0.59	+0.28	+0.27	+0.24
	smooth_fraction	−0.32	−0.29	−0.03*	−0.02*	+0.31
	stellar_mass	+0.03*	+0.02*	−0.03*	−0.07*	+0.22
CLIP-B	mag_r	−0.28	−0.35	+0.28	+0.29	+0.31
	photo_z	−0.41	−0.40	+0.08*	+0.06*	+0.25
	smooth_fraction	−0.22	−0.24	+0.15	+0.20	+0.28
	stellar_mass	−0.14	−0.13	−0.04*	+0.01*	+0.19
ConvNeXt-B	mag_r	−0.36	−0.34	+0.19	+0.21	+0.29
	photo_z	−0.46	−0.45	+0.18	+0.19	+0.30
	smooth_fraction	−0.15	−0.19	−0.02*	+0.03*	+0.32
	stellar_mass	−0.05*	−0.01*	+0.14	−0.03*	+0.29
ViT-L	mag_r	−0.46	−0.49	+0.09*	+0.15	+0.30
	photo_z	−0.44	−0.42	+0.09*	+0.13	+0.24
	smooth_fraction	−0.03*	−0.04*	+0.11	+0.07*	+0.30
	stellar_mass	−0.06*	+0.01*	−0.04*	−0.03*	+0.21

Table 6: Cross-fitted mismatch and alignment partials per encoder (multi-scale control) and split-half reliability of the label Hessian.

run	label	decoder second-order term					random normal direction			
		help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	t^*	help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$
ViT-B, $d = 16$	mag_r	0.97	1.00	+0.030	−0.051	2.3	0.28	0.71	−0.006	−0.006
	photo_z	0.99	0.99	+0.028	−0.038	3.0	0.38	0.70	−0.003	−0.004
	smooth_fraction	0.90	1.00	+0.021	−0.040	1.7	0.24	0.79	−0.008	−0.009
	stellar_mass	0.96	0.99	+0.026	−0.040	2.6	0.34	0.68	−0.004	−0.005
ViT-B, $d = 20$	mag_r	0.99	1.00	+0.031	−0.049	2.6	0.30	0.72	−0.005	−0.006
	photo_z	0.99	1.00	+0.029	−0.038	3.6	0.37	0.68	−0.002	−0.004
	smooth_fraction	0.90	1.00	+0.020	−0.035	1.9	0.26	0.77	−0.007	−0.008
	stellar_mass	0.97	1.00	+0.027	−0.038	3.1	0.31	0.72	−0.003	−0.004
DINOv3, $d = 16$	mag_r	0.92	0.99	+0.018	−0.031	2.0	0.36	0.70	−0.003	−0.004
	photo_z	0.96	0.99	+0.020	−0.030	2.6	0.36	0.62	−0.002	−0.002
	smooth_fraction	0.93	1.00	+0.024	−0.055	1.5	0.28	0.72	−0.009	−0.008
	stellar_mass	0.95	1.00	+0.016	−0.028	2.1	0.32	0.64	−0.002	−0.002
CLIP-B, $d = 16$	mag_r	0.95	0.99	+0.018	−0.031	2.2	0.34	0.66	−0.004	−0.004
	photo_z	0.99	1.00	+0.019	−0.026	3.4	0.39	0.62	−0.002	−0.002
	smooth_fraction	0.92	0.99	+0.014	−0.027	1.7	0.28	0.69	−0.004	−0.004
	stellar_mass	0.97	1.00	+0.020	−0.028	2.8	0.36	0.65	−0.002	−0.002
ConvNeXt-B, $d = 16$	mag_r	0.92	1.00	+0.018	−0.031	2.0	0.32	0.64	−0.004	−0.003
	photo_z	0.99	1.00	+0.018	−0.026	2.9	0.36	0.59	−0.002	−0.001
	smooth_fraction	0.88	0.98	+0.016	−0.036	1.6	0.31	0.69	−0.004	−0.004
	stellar_mass	0.99	1.00	+0.022	−0.032	3.1	0.38	0.57	−0.002	−0.001
ViT-L, $d = 16$	mag_r	0.96	0.99	+0.022	−0.034	2.6	0.32	0.69	−0.003	−0.004
	photo_z	0.97	0.99	+0.018	−0.025	3.2	0.40	0.61	−0.002	−0.002
	smooth_fraction	0.79	0.97	+0.013	−0.029	1.6	0.24	0.77	−0.005	−0.007
	stellar_mass	0.98	1.00	+0.022	−0.032	2.9	0.35	0.64	−0.002	−0.002

Table 7: Counterfactual scaling of the readout’s in-sphere normal component, 512 anchors per run. Bending as the manifold does helps at nearly every anchor, the mirror-image bending hurts at nearly every anchor, a random normal direction does neither; $t^* > 1$ throughout, so the ridge-shrunk normal component undershoots.